Velocity graphs and components

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

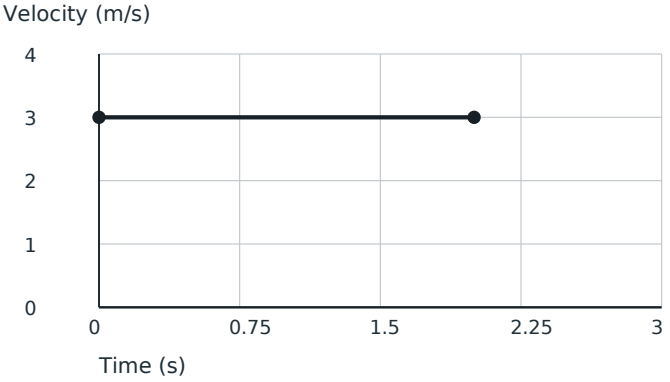
Learning target: Read a velocity graph and describe motion in two directions.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 1.*
52 m if the 6 m/s rightward motion continues. Check a later timed position against the prediction.

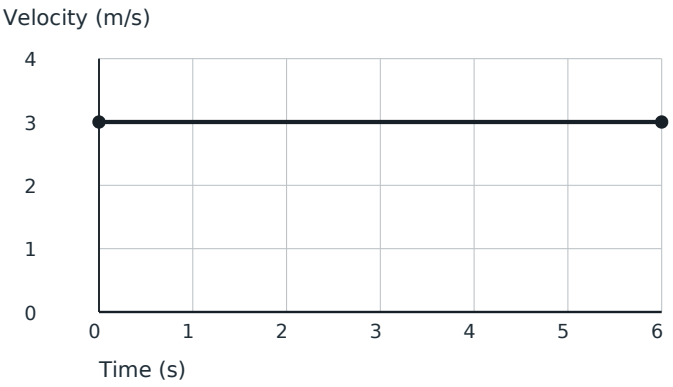
02. Worked example: explain the model
Rectangle height 3 m/s, width 2 s; area 6 m east.

Model / diagram check



03. Steady does not mean stopped
It moves steadily at +3 m/s. Displacement is the rectangle area: (3 m/s)(6 s) = +18 m.

Model / diagram check



04. Move, then wait
6 m west; waiting adds zero displacement and does not undo the trip.

05. One diagonal motion
4 m/s east and 3 m/s north. Both components occur simultaneously; the actual motion is diagonal.

06. Make the components visible
Diagonal from (0, 0) to (3, 2); components 3 m/s east and 2 m/s north.

07. Fix the speed claim
Components are perpendicular, not along one line. The diagonal is 5 m/s as supplied; 4+3 is not its magnitude. No square-root calculation required.

08. Review: connect an earlier idea *Review: Unit 1, Day 3.*
v=(5-15)/3=-3.333 m/s, left. Straight descending graph between the stated endpoints.

09. Exit check
Displacement -10 m, or 10 m in the negative direction. Second sketch points southeast, with east component twice the south component.

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10. STRETCH: test the model
Example +2 m/s for 2 s then -2 m/s for 2 s: areas cancel, distance 8 m.

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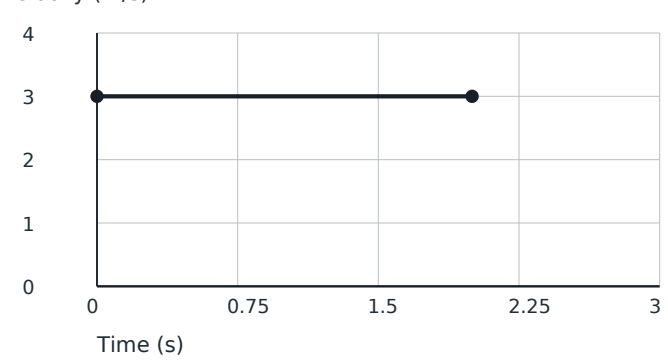
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Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 1.*
v=-3 m/s; x(8)=-6 m. Assumes constant velocity; compare another timed position with its uncertainty.

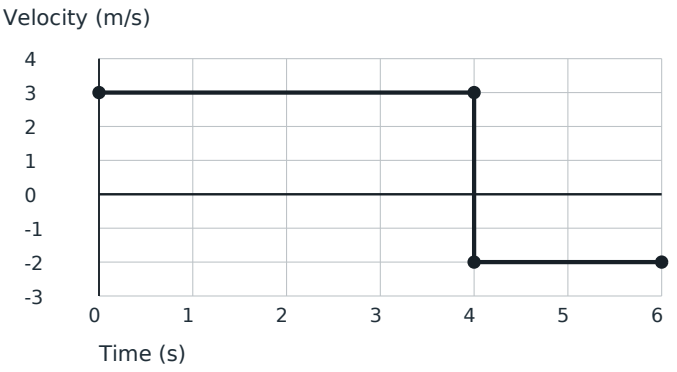
02. Worked example: explain the model
Rectangle height 3 m/s, width 2 s; area 6 m east.

Model / diagram check



03. Add signed areas
First 4 s: +12 m. Last 2 s: -4 m. Displacement +8 m; distance 16 m. Motion positive then negative.

Model / diagram check



04. Resolve a northeast velocity
v_x=14 cos 35°=11.5 m/s; v_y=14 sin 35°=8.03 m/s. East component larger because the direction is closer to east.

05. Change the reference axis
v_x=-10 sin 20°=-3.42 m/s; v_y=10 cos 20°=9.40 m/s. North component is adjacent to the stated angle.

06. Draw velocity and displacement
Speed 5 m/s. Displacement (8,-6) m, magnitude 10 m, same southeast direction.

07. Two horizontal graphs
x=5 m: stationary at that position, v=0. v=5 m/s: constant positive motion with zero acceleration; position changes 5 m each second.

08. Review: connect an earlier idea *Review: Unit 1, Day 3.*
v=5 m/s; position increases 5 m each second. x(7)=27 m.

09. Exit check
Displacement 6-2=4 m; distance 6+2=8 m. Need reference axis and direction/sense of angle.

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10. STRETCH: test the model
Positive displacement 12 m; return takes 6 s. Distance 24 m in 9 s gives 8/3 m/s; average velocity zero.